Kengo Tanaka

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RESEARCH AREAS

Accessibility · Human–computer interaction · Tactile graphics · Embodied interaction · Inclusive sports

EDUCATION

University of Tsukuba, Japan Apr 2024 – Present
Doctor of Informatics (in progress), Digital Nature Group
Advised by Professor Yoichi Ochiai

University of Tsukuba, Japan Apr 2021 – Mar 2023
Master of Informatics, Digital Nature Group
Advised by Associate Professor Yoichi Ochiai

University of Tsukuba, Japan Apr 2017 – Mar 2021
Bachelor of Science in Media Sciences and Engineering, Digital Nature Group
Advised by Associate Professor Yoichi Ochiai

RESEARCH & PROFESSIONAL APPOINTMENTS

JSPS Research Fellow (DC2) Apr 2026 – Present
Research Fellowships for Young Scientists, Japan Society for the Promotion of Science (JSPS)

Part-time Researcher · Miraikan, National Museum of Emerging Science and Innovation Mar 2025 – Present

JST BOOST recipient Apr 2025 – Mar 2026
Broadening Opportunities for Outstanding Young Researchers and Doctoral Students in Strategic Areas (Next-Generation AI Human Resource Development Program), Japan Science and Technology Agency (JST)

Advertising production company Apr 2023 – Sep 2024

PUBLICATIONS

Journal Papers

Kengo Tanaka, Xiyue Wang, Hironobu Takagi, Yoichi Ochiai, Chieko Asakawa. 2026. OdoriCo: 3D-Printed Pose Sequences on a Rotating Table for Blind Learners. The Journal of the Society for Art and Science Vol. 25, No. 2, pp. 13:1–13:14. https://www.art-science.org/journal/v25n2/v25n2-13/artsci-v25n2-13.pdf

Conference Full Papers

Kengo Tanaka, Renato Alexandre Ribeiro, João Guerreiro, Hironobu Takagi, Yoichi Ochiai, and Chieko Asakawa. 2026. From Coarse Search to Precise Alignment: Auditory Guidance for Non-Visual Target Acquisition in VR. In The 28th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '26), October 25–28, 2026, Vila Nova de Gaia, Portugal. ACM, New York, NY, USA, 16 pages. https://doi.org/10.1145/3797867.3828980

Kengo Tanaka, Xiyue Wang, Hironobu Takagi, Yoichi Ochiai, and Chieko Asakawa. 2026. MotionBuddy: Exploring Tactile-Based Motion Learning with a Tabletop Humanoid Robot for Blind People. In Proceedings of the 21st ACM/IEEE International Conference on Human-Robot Interaction (HRI '26). Association for Computing Machinery, New York, NY, USA, 608–618. https://doi.org/10.1145/3757279.3788660

Kengo Tanaka, Xiyue Wang, Hironobu Takagi, Yoichi Ochiai, and Chieko Asakawa. 2026. Touching Movement: 3D Tactile Poses for Supporting Blind People in Learning Body Movements. In Proceedings of the Twentieth International Conference on Tangible, Embedded, and Embodied Interaction (TEI '26). Association for Computing Machinery, New York, NY, USA, Article 29, 1–19. https://doi.org/10.1145/3731459.3773325

K. Tanaka, A. Tsutsui, T. Murakami and Y. Ochiai, "Co-Creating Tactile Graphics from Picture-Book Illustrations with Generative AI," 2026 Nicograph International (NICOInt), Nagoya, Aichi, Japan, 2026, pp. 38–45, doi: 10.23919/NICOIntCPS00076.2026.00019.

Ayaka Tsutsui, Kenta Yamamoto, Yinan Zhao, Ippei Suzuki, **Kengo Tanaka**, and Yoichi Ochiai. 2024. Low Vision Boxing: Participatory Design of Adaptive Kickboxing Experiences with Low Vision Person. In Proceedings of the 26th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '24). Association for Computing Machinery, New York, NY, USA, Article 15, 1–18. https://doi.org/10.1145/3663548.3675619

Ayaka Tsutsui, Tatsuki Fushimi, Takahito Murakami, Ryosei Kojima, **Kengo Tanaka**, Yoichi Ochiai. "HIFU Embossment of Acrylic Sheets", In Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI'24). https://doi.org/10.1145/3613904.3642890

Posters & Short Papers

Yoichiro Katayama, Ayaka Tsutsui, **Kengo Tanaka**, Tatsuki Fushimi, and Yoichi Ochiai. 2026. Toward Automated Commentary System for Blind Football: An Interview-Based Study of Visually Impaired Spectators' Information Needs. In Proceedings of the 2026 28th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '26), 1–5. https://doi.org/10.1145/3797867.3841225

Ayaka Tsutsui, Takahito Murakami, Tatsuki Fushimi, Ryosei Kojima, **Kengo Tanaka**, and Yoichi Ochiai. 2026. Evaluating Tactile Perception of Robot-Controlled HIFU Embossing. In Companion Proceedings of the 21st ACM/IEEE International Conference on Human-Robot Interaction (HRI Companion '26). Association for Computing Machinery, New York, NY, USA, 558–562. https://doi.org/10.1145/3776734.3794457

Kengo Tanaka, Ayaka Tsutsui, and Yoichi Ochiai. 2025. A Comparative Evaluation of GenAI-Assisted Tactile Graphics from the Perspective of Totally Blind Professionals in Tactile Graphic Supervision. In Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '25). Association for Computing Machinery, New York, NY, USA, Article 147, 1–5. https://doi.org/10.1145/3663547.3759740

Ayaka Tsutsui, **Kengo Tanaka**, Yaman Yu, Bektur Ryskeldiev. "I Am a Blind Seller!": Picture Taking Assistance for Visually Impaired Individuals for Participation as Sellers in Customer to Customer (C2C) Marketplaces. DOI: https://doi.org/10.1145/3706599.3720150

Masaaki Sadasue, Ippei Suzuki, Kosaku Namikawa, **Kengo Tanaka**, Chieko Nishimura, Masaki Okamura, Yoichi Ochiai. (2022). Development and Evaluation of Systems to Enjoy a Wedding Reception for People with Low Vision. In: Stephanidis, C., Antona, M., Ntoa, S. (eds) HCI International 2022 Posters. HCII 2022. Communications in Computer and Information Science, vol 1580. Springer, Cham. DOI: https://doi.org/10.1007/978-3-031-06417-3_79

Kohei Ogawa, **Kengo Tanaka**, Tatsuya Minagawa, and Yoichi Ochiai. 2018. Design method of digitally fabricated spring glass pen. In ACM SIGGRAPH 2018 Poster (SIGGRAPH '18). ACM, New York, NY, USA, Article 3, 2 pages. DOI: https://doi.org/10.1145/3230744.3230809

Kengo Tanaka, Taisuke Ohshima, and Yoichi Ochiai. 2017. Spring-pen: reproduction of any softness with the 3D printed spring. In SIGGRAPH Asia 2017 Posters (SA '17). ACM, New York, NY, USA, Article 37, 2 pages. DOI: https://doi.org/10.1145/3145690.3145732

Preprints

Vargas Meza, Xanat & **Tanaka, Kengo** & Ochiai, Yoichi. (2025). Transnational Ainu media: YouTube Itak before and after 2019. 10.21203/rs.3.rs-8265421/v1.

Ryosei Kojima, Akihisa Shitara, Tatsuki Fushimi, Ryogo Niwa, Atushi Shinoda, Ryo Iijima, **Kengo Tanaka**, Sayan Sarcar, and Yoichi Ochiai. 2023. SHITARA: Sending Haptic Induced Touchable Alarm by Ring-shaped Air vortex. arXiv:2301.08107 [cs.HC]. https://doi.org/10.48550/arXiv.2301.08107

Demos & Studio

Kengo Tanaka, Tatsuki Fushimi, Ayaka Tsutsui, and Yoichi Ochiai. "Text to Haptics: Method and Case Studies of Designing Tactile Graphics for Inclusive Tactile Picture Books by Digital Fabrication and Generative AI", In Special Interest Group on Computer Graphics and Interactive Techniques Conference Labs (SIGGRAPH '23 Labs). https://doi.org/10.1145/3588029.3595471

Ayaka Tsutsui, Tatsuki Fushimi, **Kengo Tanaka**, Takahito Murakami, Yoichi Ochiai. "Ultrasonic Embossment of Acrylic Sheets with Transparency Control", In Special Interest Group on Computer Graphics and Interactive Techniques Conference Labs (SIGGRAPH '23 Labs). https://doi.org/10.1145/3588029.3595475

Kengo Tanaka, Kohei Ogawa, Tatsuya Minagawa, and Yoichi Ochiai. 2018. Design method of digitally fabricated spring glass pen. In ACM SIGGRAPH 2018 Studio (SIGGRAPH '18). ACM, New York, NY, USA, Article 3, 2 pages. DOI: https://doi.org/10.1145/3214822.3214825

Video Showcase

Kenta Yamamoto, Ayaka Tsutsui, Yinan Zhao, Ippei Suzuki, **Kengo Tanaka**, Yusuke Koroyasu, and Yoichi Ochiai. 2024. Documentary Film of Low Vision Boxing: Collaborative Design of Adaptive Kickboxing Experiences. In Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24). Association for Computing Machinery, New York, NY, USA, Article 641, 1–2. https://doi.org/10.1145/3613905.3649124

GRANTS & FELLOWSHIPS

2026	JSPS Research Fellowships for Young Scientists (DC2) · JPY 2.8 million / 2 years
2026	30th JRPS Research Grant, Lions Award · JPY 1 million “Support for Movement and Spatiotemporal Awareness of People with Low Vision Using Generative AI”
2026	AI for Science (SPReAD), second call · JPY 3 million “Developing a Brushstroke Physical AI Platform for Calligraphy Learning by People with Visual Impairments”
2026	Univers Foundation Research Grant · JPY 0.8 million Support for Exploring Exhibits Behind Glass in Museums and Science Centers for Visitors with Visual Impairments
2026	Cyber-Medicine Young Researcher Support (FY2025, 2nd call) · JPY 1 million Exploratory Study of Facial and Vocal Indicators for AI-Based Mental Health Support for Postpartum Women
2026	Tsukuba Marathon Science Support Program · JPY 0.2 million “Investigating AI-Based Scenery Sharing for Runners with Visual Impairments”
2025	Tsukuba Smart City Social Implementation Trial Support Program · JPY 1 million “Illuminated Kickboxing Accessible to People with Visual Impairments”
2025	JST BOOST, Next-Generation AI Human Resource Development Program · JPY 3 million / 2 years

AWARDS & ACHIEVEMENTS

1. **Excellence Award**; NYAA 2024 (Next Young Artist Award), Game & Interaction Division, HIFU Embossment — Mar 2025
2. **Grand Prize**; 29th Campus Genius Contest, Low Vision Boxing (0.66%; 3 of 451) — Feb 2024
3. **2nd Prize**; Creative Hack Award 2023, Low Vision Boxing (0.9%; 2 of 220) — Dec 2023
4. **Accenture Interactive Award**; 27th Campus Genius Contest, Drone Badminton (6.1%; 26 of 429) — Apr 2022
5. **Dean's Award**; School of Information Sciences, University of Tsukuba (4.8%; 3 of 63) — Mar 2021
6. **Special Prize**; enPiT 2019 Agile Development Course, University of Tsukuba (21%; 10 of 47) — Feb 2020
7. **Special Prize**; ARE 2017 (Advancing Researcher Experience), Spring-Pen (21%; 10 of 47) — Feb 2018
8. **Finalist**; Creative Hack Award 2017, Spring-Pen (4.1%; 18 of 437) — Feb 2018
9. **Student Award**; YouFab Global Creative Awards 2017, Spring-Pen (2.2%; 5 of 227) — Feb 2018
10. **Nominated**; 23rd Campus Genius Contest, Spring-Pen (21.5%; 87 of 404) — Feb 2018

SELECTED TECHNICAL SKILLS

Programming & web: Python, C/C++, JavaScript (Next.js), Swift, HTML, CSS
XR development: Unity, XR interaction prototyping
Backend & database: Firebase
Adobe: Premiere Pro, After Effects, Photoshop, Illustrator, XD
Analytics: Google Analytics 4 (GA4)
Digital fabrication: Laser cutter, 3D printing (FDM, SLA, SLS)
Hardware: Arduino, Raspberry Pi
3D CAD & modeling: Fusion 360, Blender